

# Hardest tech we ever tried yet

**O**n paper, India's semiconductor ambition is staggering. The India Semiconductor Mission now has 10 approved projects across six states, representing investments of roughly ₹1.6 lakh crore and backed by a ₹76,000-crore incentive framework offering up to 50% fiscal support for fabs, packaging units and design infrastructure. The first commercial chip production under this push is already beginning in 2026, with multiple fabrication and packaging facilities expected to follow.

But spend enough time speaking to people in the know, and a far more nuanced narrative begins to emerge.

One senior industry veteran described India's semiconductor talent pipeline using an analogy that has stayed with me. Our premier institutes, he said, are excellent at teaching the "ABCs" of semiconductors. But the industry doesn't need alphabet-level knowledge anymore. It needs engineers who can speak in phrases, maybe even complete sentences. People who can be trained to write the full story.

Right now, even top-tier graduates still require months of intensive industry training before they become productive. The good news? Once that bridge is built, they scale quickly. The uncomfortable truth? Industry needs to invest far more deeply and far earlier with academia if India wants a workforce that can keep pace with the fabs and design centres now being planned.

That workforce challenge sits alongside a second, more philosophical realisation. India isn't yet on the global semiconductor playing field – even now, we're still warming up while the match is already in full swing.

As one global semiconductor executive told me, chipmaking isn't an industry where you win once and celebrate. It's a game where the score keeps changing and the players keep evolving – and you only get influence once you're actually in the match. Countries don't negotiate their place in this ecosystem from a standing start – they negotiate from dependable participation.



**“A billion dollars spent on fabs could transform strategic capability – or it could have been spent on improving basic infrastructure for millions of Indians.”**

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Right now, India is still stepping onto that field. The real question isn't whether we can dominate every segment. No country can. The real question is: which bargaining chips will India build over time: Materials? Equipment? Packaging? AI-driven chip design? Process innovation? Nobody can predict which of these will become decisive. Innovation rarely fits neatly into government spreadsheets.

This unpredictability makes policymaking particularly tricky. Semiconductor investments are enormous, long-gestation bets. A billion dollars spent on fabs could transform strategic capability – or it could have been spent on improving basic infrastructure for millions. For any government, especially in a developing nation, that trade-off is brutally real.

And yet, amid the caution, there's also quiet optimism from within the ecosystem.

Several industry insiders spoke of an unexpected level of alignment between government and industry – sometimes cutting across political lines. One executive narrated how regulatory roadblocks around building an R&D facility were resolved in mere days after escalation to the right ministries. Another spoke of bureaucrats diving two technical layers deeper into packaging technology discussions than anyone expected.

These aren't just anecdotes, I think they hint at something more profound – of a growing institutional seriousness about semiconductors that goes beyond headline announcements.

Which brings us to the bigger picture. India's semiconductor journey isn't just about fabs or incentives or global partnerships. It's about whether the country can stake a meaningful claim in at least one or two critical layers of the global value chain – and then build outward from there.

Because in semiconductors, relevance compounds slowly. One capability leads to another. One facility attracts an ecosystem. One success builds negotiating leverage.

India doesn't need to win the entire race. Not yet. It just needs to get onto the field – and stay in the game enough to matter. **d**